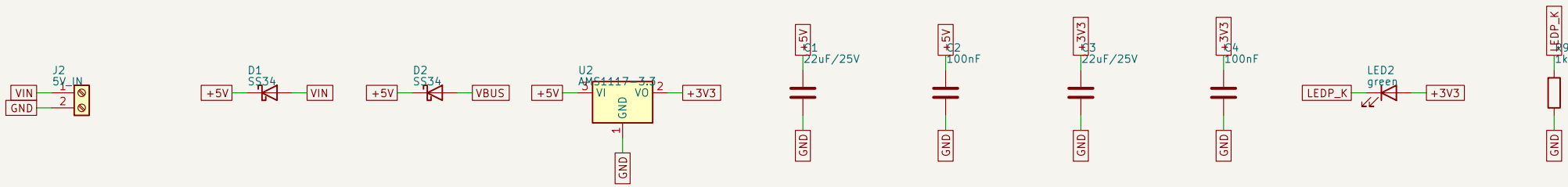
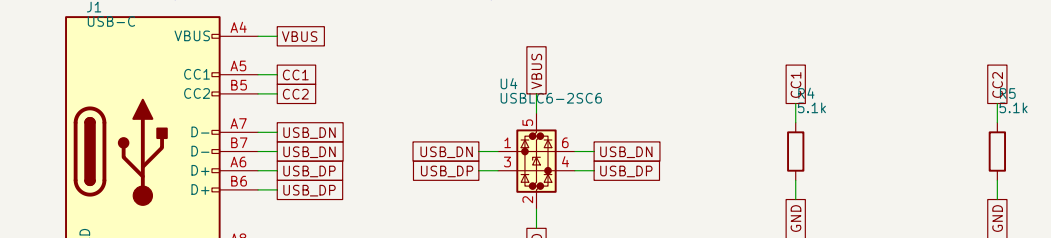


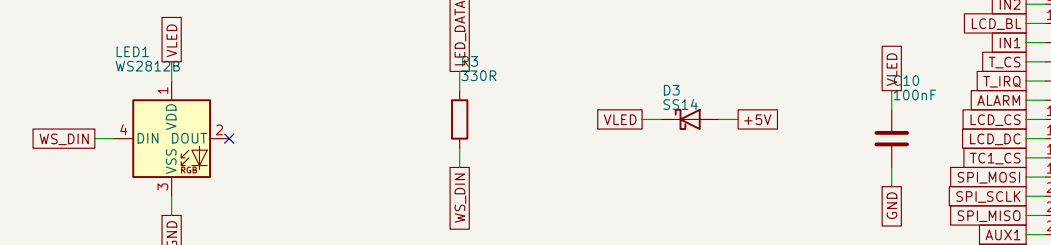
POWER IN (5V DC terminal or USB, Ored Schottky diodes)



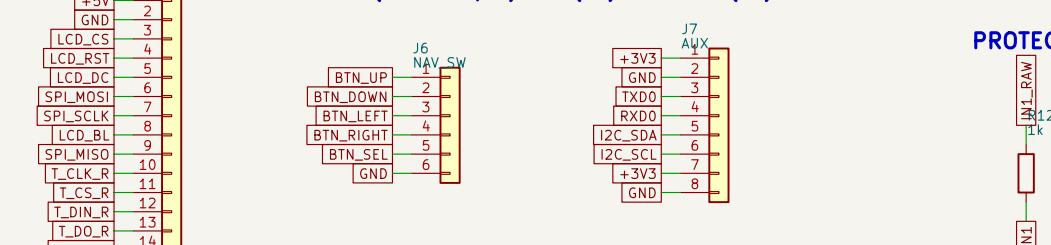
USB-C (native USB flashing + ESD)



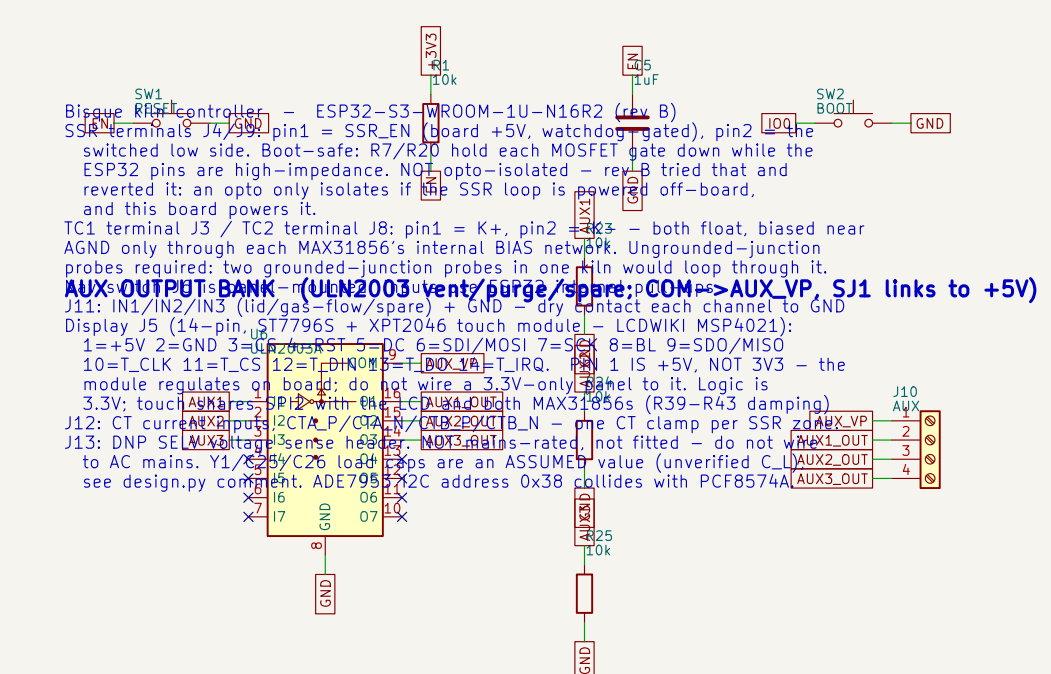
WS2812B STATUS LED (VDD dropped -4.6V for 3.3V data margin)



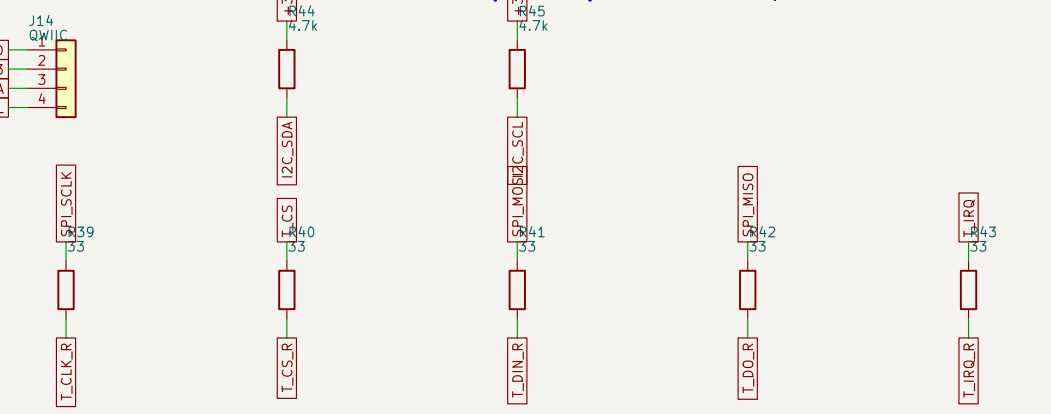
HEADERS: DISPLAY+TOUCH(J5, 14-pin) NAV(J6) AUX+I2C(J7)



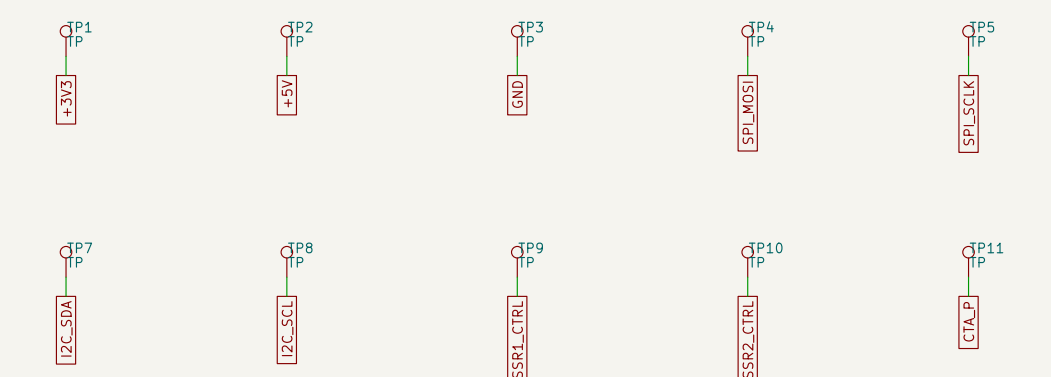
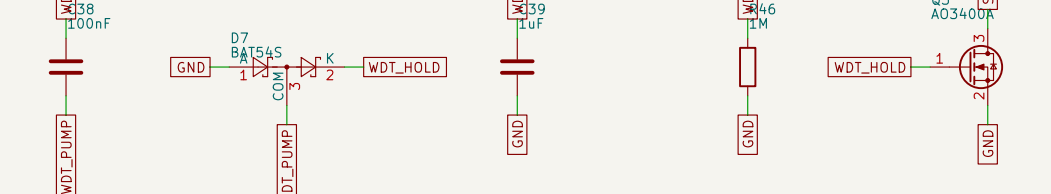
RESET / BOOT / DECOUPLING



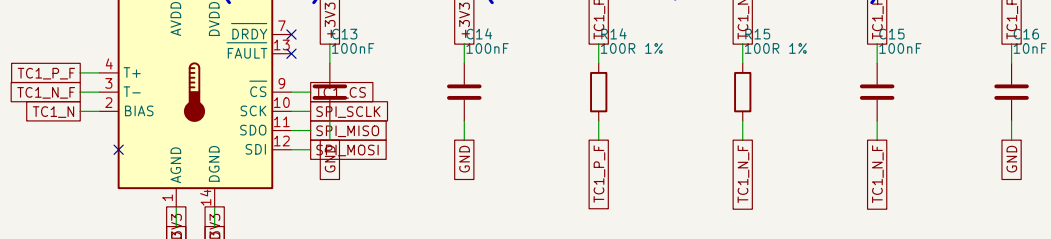
TOUCH DAMPING + I2C EXPANSION (Task 11) R39-43 damp the shared SPI2nbus for the display module's XPT2046 (not on this board); J14 Qwilk +nJ7 5-8 (0.1") share the I2C bus, pulled up by R44/R45



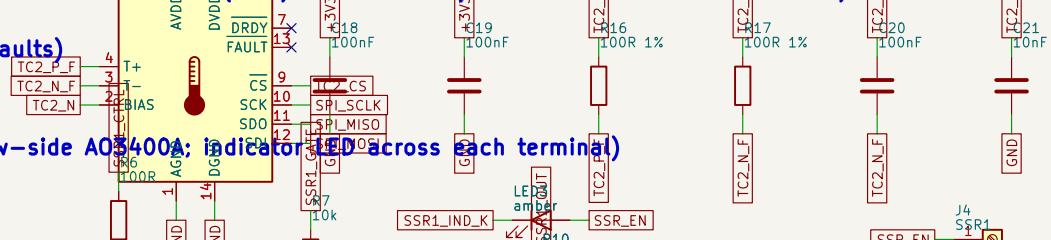
HARDWARE WATCHDOG C38/D7/C39/R46 diode charge pump on WDT_KICK holds nQ3 on; Q3 pulls SSR_PG down, turning on high-side Q4, which supplies nSSR_EN - the +5V rail feeding BOTH SSR terminals and both indicators. nR47 is the fail-safe pull-up. S12 = bring-up defeat. REMOVE for service.



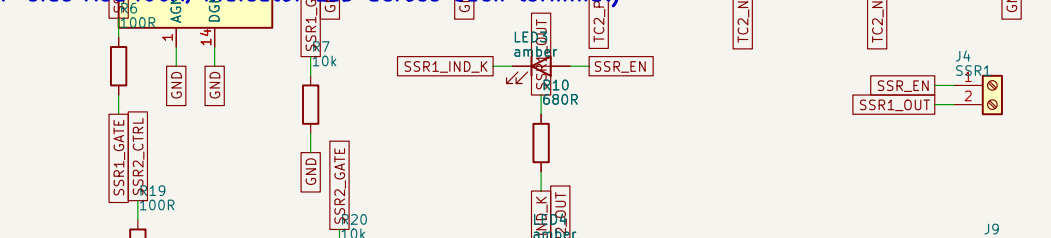
THERMOCOUPLE 1 (control) MAX31856 (T- floats to J3, biased via BIAS)



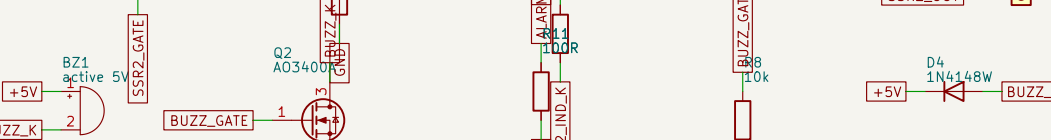
THERMOCOUPLE 2 (load) MAX31856 (T- floats to J8, biased via BIAS)



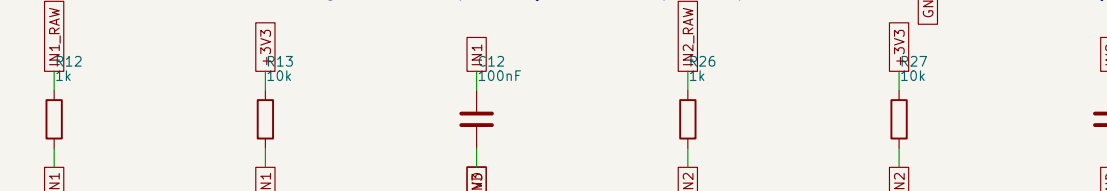
SSR DRIVE x2 (low-side A08400A: indicator LED across each terminal)



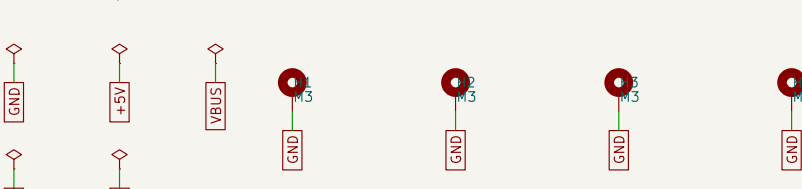
ALARM BUZZER



PROTECTED INPUTS x3 lid/gas-flow/spare (1k + 10k pull-up + 100nF each; SRV05-4 TVS)



MOUNTING / POWER FLAGS



CT CURRENT SENSING ADE7953, I2C, current-only (no mains - VP/VN to DNP J13)

